MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

13

10 answers · Rationale-rich · Teach from doc

Q1

A

A. $k = \frac{m - 14j}{5}$

Choice A is correct. Subtracting $14j$ from each side of the given equation results in $5k = m - 14j$. Dividing each side of this equation by 5 results in $k = \frac{m - 14j}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

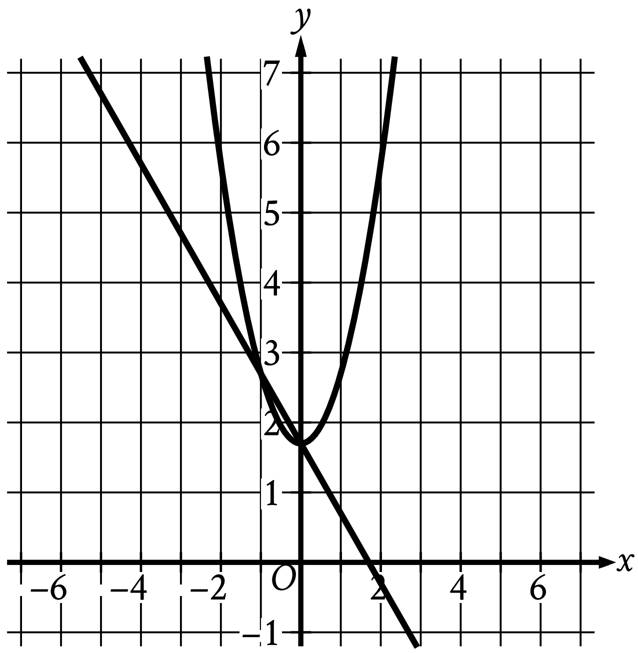
Q2

2

The correct answer is 2.5. After hitting the ground once, the ball bounces to $20.0 \div 2 = 10.0$ feet. After hitting the ground a second time, the ball bounces to $10.0 \div 2 = 5.0$ feet. After hitting the ground for the third time, the ball bounces to $5.0 \div 2 = 2.5$ feet. Note that 2.5 and 5/2 are examples of ways to enter a correct answer.

Q3

A



A.

- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

Choice A is correct. The graph of a quadratic equation in the form $y = x^2 + c$ has its vertex at $(0, c)$. The first equation in the given system of equations is $y = x^2 + 1.7$, so the graph of this quadratic equation has its vertex at $(0, 1.7)$. The graph of a linear equation of the form $y = b - x$ has a slope of -1 and a y-intercept at $(0, b)$. The second equation in the given system of equations is $y = 1.7 - x$, so the graph of this linear equation has a slope of -1 and a y-intercept at

$(0, 1.7)$. Of the choices, only choice A has the graph of a quadratic equation with its vertex at $(0, 1.7)$ and the graph of a linear equation with a slope of -1 and a y-intercept at $(0, 1.7)$.

Choice B is incorrect. This graph represents a system in which the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Choice C is incorrect. This graph represents a system in which the first equation is $y = \left(x + 1.7\right)^2$, not $y = x^2 + 1.7$.

Choice D is incorrect. This graph represents a system in which the first equation is $y = \left(x + 1.7\right)^2$, not $y = x^2 + 1.7$, and the second equation is $y = 1.7 + x$, not $y = 1.7 - x$.

Q4

B

B. The predicted body mass of the animal was approximately 362 kg 330 days after it was born.

Choice B is correct. In the statement "$m(330)$ is approximately equal to 362," the input of the function, 330, is the value of t, the elapsed time, in days, since the animal was born. The approximate value of the function, 362, is the predicted body mass, in kilograms, of the animal after that time has elapsed. Therefore, the predicted body mass of the animal was approximately 362 kg 330 days after it was born.

Choice A is incorrect. This would be the best interpretation of the statement "$m(362)$ is approximately equal to 330."

Choice C is incorrect. The number $\frac{330}{7}$ is the number of weeks, not the number of days, after the animal was born.

Choice D is incorrect. This would be the best interpretation of the statement "$m(362)$ is approximately equal to $\frac{330}{7}$."

Q5

C

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

Choice C is correct. Using the quadratic formula to solve the given expression yields $x = \frac{-(-1) \pm \sqrt{(-1)^2 - (4)(1)(-1)}}{(2)(1)} = \frac{1 \pm \sqrt{5}}{2}$. Therefore, $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$ satisfy the given equation.

Choices A and B are incorrect and may result from incorrectly factoring or incorrectly applying the quadratic formula. Choice D is incorrect and may result from a sign error.

Q6

B

B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice B is correct. On the graph shown, the y-axis represents estimated population, in thousands. The graph shows that when $x = 0$, the y-coordinate is 6. Therefore, the estimated population at $x = 0$ is 6 thousand. The graph also shows that when $x = 1$, the y-coordinate is 9. Therefore, the estimated population at $x = 1$ is 9 thousand. Dividing 9 thousand by 6 thousand yields 1.5; therefore, 9 thousand is 1.5 times 6 thousand. It follows that the estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

either 8 or 9

The correct answer is either 8 or 9. The first equation can be rewritten as $y=17-x$. Substituting $17-x$ for y in the second equation gives $x(17-x)=72$. By applying the distributive property, this can be rewritten as $17x-x^2=72$. Subtracting 72 from both sides of the equation yields $x^2-17x+72=0$. Factoring the left-hand side of this equation yields $(x-8)(x-9)=0$. Applying the Zero Product Property, it follows that $x-8=0$ and $x-9=0$. Solving each equation for x yields $x=8$ and $x=9$ respectively. Note that 8 and 9 are examples of ways to enter a correct answer.

Q8

A

A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.

Choice A is correct. In the given function, st represents the approximate mass, in grams, of the sample that remains t years after the sample starts to decay. It follows that the best interpretation of $s87 = 23$ is that approximately 23 grams of the sample remains 87 years after the sample starts to decay.

Choice B is incorrect. The mass of the sample has decreased by approximately 184 - 23, or 161, grams, not 23 grams, 87 years after the sample starts to decay.

Choice C is incorrect. The mass of the sample has decreased by approximately 78 grams, not 87 grams, 23 years after the sample starts to decay.

Choice D is incorrect. This would be the best interpretation of $s23 = 87$, not $s87 = 23$.

Q9 C

C. 42

Choice C is correct. Multiplying both sides of the given equation by x yields $42 = 7x^2$. Therefore, the value of $7x^2$ is 42.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 68

The correct answer is 68. It's given that the function f is defined by $fx = 8x^3 + 4$. Substituting 2 for x in this equation yields $f2 = 82^3 + 4$, or $f2 = 88 + 4$, which is equivalent to $f2 = 68$. Therefore, the value of $f2$ is 68.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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